Symbolic Health Equations for Unsupervised Gearbox Fault Detection via Kolmogorov–Arnold Autoencoders

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***Abstract*—Most deep anomaly detectors for rotating machinery are black boxes: they flag a fault but cannot say *which physical relationship* broke down. We propose a framework that converts a Kolmogorov–Arnold Network (KAN) autoencoder—trained on healthy gearbox vibration data only—into a set of closed-form "health equations" that make fault detection interpretable by design. Each learned activation is approximated by a simple mathematical primitive, yielding 22 explicit equations that characterise normal operation. At test time, deviations from these equations serve as anomaly scores: a Mahalanobis distance on equation residuals achieves AUC 0.9891 ± 0.0069 (mean $\pm$ std, $N{=}10$ seeds) on the SpectraQuest broken-tooth benchmark, surpassing SHAP-weighted reconstruction (0.9535 ± 0.0129) and all deep baselines while remaining second only to LOF among all eighteen methods compared. A fully symbolic variant—which discards the neural network entirely at inference—runs in 0.018 ms per sample ($200\times$ faster than SHAP) and still attains AUC 0.9777 ± 0.0075. Critically, violated equations attribute faults to specific sensor-level statistics (e.g., S1_rms, S1_kurtosis) rather than returning a global feature ranking as SHAP and LIME do.**

***Index Terms*—Kolmogorov–Arnold networks, symbolic regression, interpretable anomaly detection, gearbox fault diagnosis, vibration analysis, unsupervised learning.**

I. Introduction

Gearbox failures account for a disproportionate share of unplanned downtime in rotating machinery. Broken-tooth faults are particularly insidious: they propagate rapidly under load, can cause catastrophic drivetrain damage within seconds, and yet are extremely difficult to detect before they manifest as vibration anomalies [1].

The standard diagnostic paradigm—train a supervised classifier on matched healthy/faulty recordings—is almost never achievable in practice. Faults are intentionally rare events; inducing them for data collection destroys the equipment under test; and the vibration signature of a given fault depends heavily on load level, shaft speed, and machine age, requiring exhaustive experimental coverage. Practitioners are left with abundant healthy operational data and almost no labeled fault data.

Condition-based maintenance of industrial gearboxes relies on vibration analysis to catch incipient failures before catastrophic breakdown. Modern data-driven detectors—autoencoders, variational autoencoders, one-class classifiers, patch-based memory networks—achieve near-perfect ROC-AUC on benchmark datasets, but they share a common shortcoming: they produce a *scalar anomaly score with no physical interpretation*. Post-hoc explainers such as SHAP or LIME partially remedy this by assigning per-feature importance weights, yet (i) their explanations require running a surrogate model at inference, (ii) they are global averages that do not identify *which structural relationship* was violated, and (iii) they cannot be deployed without the original neural network.

Kolmogorov–Arnold Networks (KANs) [2] recently reintroduced learnable B-spline activations on *edges* rather than fixed activations on nodes. Because each spline is a smooth univariate function of one feature, KANs are naturally amenable to *symbolic distillation*: every edge can, in principle, be replaced by a human-readable mathematical primitive. Prior work on KANs has focused on supervised regression and classification in physics and PDE discovery [2], [3]; to our knowledge, none has exploited KAN splines for *unsupervised symbolic health modelling* of rotating machinery.

This paper introduces the following novel contributions :

1) **Symbolic health equations from KAN autoencoders.** We show that a KAN autoencoder trained only on healthy vibration features learns per-edge activations that are almost perfectly symbolisable—100% of surviving edges reach $R^2 > 0.95$ (coefficient of determination) against a library of thirteen symbolic primitives, with mean $R^2 = 0.9999$. The resulting closed-form equations describe the healthy operating manifold of a gearbox with four accelerometer channels.
2) **Multiple symbolic anomaly scores.** We demonstrate that calibrated symbolic-violation and Mahalanobis-on-symbolic-residuals scores yield fully interpretable detectors (mean AUC $0.9026{\pm}0.0558$ and $0.9891{\pm}0.0069$ respectively over $N{=}10$ seeds). Furthermore, by symbolising the entire autoencoder into a stack of simple mathematical equations, we achieve a *fully symbolic reconstruction* AUC of 0.9650. This completely discards the neural network at inference time, enabling an extremely lightweight and understandable deployability. Finally, a combined reconstruction + symbolic-violation detector achieves AUC 0.9716 ± 0.0151—surpassing the opaque reconstruction score (0.9642 ± 0.0157) and the majority of unsupervised baselines including VAE, DeepSVDD, Teacher–Student distillation, and SSL-DAE.

3) **Equation-level fault localisation.** Because each symbolic equation references an explicit subset of sensor-level statistics, a *per-equation violation ratio* identifies the dominant statistical features of the damaged channel(s) with no fault labels. On the SpectraQuest benchmark the most violated equation (ratio $16.19\times$ healthy) is dominated by S1 statistics (S1_mean, S1_rms, S1_std)—features of the accelerometer physically closest to the broken tooth.
4) **Principled ablation of symbolisability.** We show that the L_1+entropy spline regulariser of `efficient-kan` is the mechanism that drives high symbolisation quality: turning it off drops quality from 100% to 98.7% and increases the surviving-edge count by 8.5%. A bottleneck sweep $B \in \{4, \ldots, 12\}$ further establishes that $B = 8$ is the sweet spot between reconstruction fidelity and parsimony.

The remainder of this paper is organised as follows. Section II reviews interpretable anomaly detection and KAN literature. Section III details the five-stage pipeline. Section IV describes the dataset and benchmark protocol. Section V presents the full results, and Section VI discusses limitations and directions for future work.

II. Related Work

A. Unsupervised Fault Detection for Rotating Machinery

Classical unsupervised detectors for vibration data include Isolation Forest, One-Class SVM and Local Outlier Factor. Deep variants—autoencoders, variational autoencoders, Deep-SVDD [10], Teacher–Student distillation [11] and the patch-memory PatchCore [7]—currently dominate public benchmarks on bearing and gearbox datasets, but their decisions remain opaque.

B. Kolmogorov–Arnold Networks

The KAN reformulation of Liu *et al.* [2] replaces fixed node non-linearities with learnable per-edge B-splines, yielding compact and interpretable function approximators. Subsequent work has extended KANs to Wav-KAN, FourierKAN and physics-informed variants, and applied them to supervised fault classification of bearings and gearboxes [3]. However, prior fault-diagnosis applications of KANs are (i) *supervised*, requiring labelled fault examples, and (ii) focus on classification accuracy rather than symbolic extraction.

C. Symbolic Regression and Model Distillation

Symbolic regression with PySR, AI-Feynman, or eureqa has been used to distil scientific laws from deep networks [18]. The novelty of the present work is to apply symbolic distillation *edge-wise* to a KAN autoencoder trained on healthy data only, producing a bank of closed-form equations that collectively constitute a *physical health model* of the machine.

`Pipeline.png`

Fig. 1. End-to-end pipeline. Healthy vibration data are mapped to 44 statistical features, compressed by a [44→22→8→22→44] KAN autoencoder, then each surviving encoder edge is distilled into a symbolic primitive. Anomalies violate both the reconstruction and the symbolic health equations.

III. Methodology

The proposed pipeline comprises five stages (Fig. 1): (1) statistical feature extraction, (2) KAN autoencoder training on healthy data only, (3) edge pruning and per-edge symbolic regression, (4) composition of health equations, and (5) dual anomaly scoring (reconstruction + symbolic residual).

A. Feature Extraction

For each accelerometer channel $S_c,\ c \in \{1, 2, 3, 4\}$ and each non-overlapping window of length W samples, we compute eleven time-domain statistics: mean, RMS, standard deviation, variance, skewness, kurtosis, peak-to-peak, crest factor, shape factor, impulse factor (similar to Hassan et al. [19]), and margin factor. This produces a feature vector $\mathbf{x} \in \mathbb{R}^{44}$ per window. Eight window sizes $W \in \{300, 400, \ldots, 1000\}$ are evaluated. Features are scaled to $[0, 1]$ with a `MinMaxScaler` fitted on the training-healthy split only to avoid leakage.

B. KAN Autoencoder Architecture

We use the `efficient-kan` implementation of the learnable-spline KAN. The autoencoder has layer widths $[44, 22, B, 22, 44]$ with grid size 5 and spline order 3; the bottleneck is fixed at $B{=}8$ after ablation (Sec. V). Training minimises the composite loss

$$\mathcal{L} = \underbrace{\|\mathbf{x} - \hat{\mathbf{x}}\|_2^2}_{\text{MSE recon.}} + \lambda\, \mathcal{R}_{\text{spline}}, \quad \lambda = 10^{-4}, \tag{1}$$

where $\mathcal{R}_{\text{spline}}$ is the L_1+entropy regulariser of `efficient-kan` that penalises non-smooth, high-curvature splines. The network is optimised with Adam (lr $= 10^{-3}$)

for 150 epochs on a single CPU core using *only* the healthy recordings.

C. Edge Pruning

For the first encoder layer (44→22) we compute a combined edge importance

$$c_{ji} = \overline{|w^{\text{spl}}_{ji,\cdot}|} + |w^{\text{base}}_{ji}|, \tag{2}$$

where w^{spl} is the scaled spline coefficient tensor and w^{base} is the residual linear weight. Edges with $c_{ji} < \alpha \max_{j',i'} c_{j'i'}$ are pruned (α=0.05). On our model 839 of 968 edges survive (86.7%), with the most important features being S1_kurt, S1_margin, S1_p2p and S4_impulse.

D. Per-Edge Symbolic Regression

For each surviving edge (j, i) we sweep input feature i over $[0, 1]$ with all other features held at their scaled midpoint, and read off the corresponding spline contribution $y_{ji}(x)$ in 200 equi-spaced points. We then fit thirteen candidate symbolic forms—linear, quadratic, sigmoid, $\tanh$, Gaussian, hinge, $\sqrt{\cdot}$, $\log$, constant, exponential, power, rational, and $\sin$—via SciPy `curve_fit` and select the winner by a two-stage BIC-based selection rule. First, the best polynomial fit among the polynomial candidates (linear, quadratic, constant) is determined. Then, any non-polynomial candidate whose R^2 exceeds the best polynomial by at least 0.05% is considered; if any qualify, the one with the highest R^2 is selected, otherwise the BIC-best polynomial is retained. The BIC is computed as

$$\text{BIC} = n \log(\text{MSE}) + k \log n, \tag{3}$$

where k is the number of free parameters. On our trained model, 824 of 839 surviving edges select the *quadratic* form; eleven prefer $\sin$; two prefer *hinge*; one prefers *sigmoid*; and one prefers $\tanh$. While the symbolic library includes 13 primitives, the dominance of quadratic forms suggests that healthy gearbox dynamics, after statistical feature extraction, are well-characterized by second-order relationships — consistent with energy-based interpretations. The overall symbolisation quality (fraction of edges with R^2>0.95) is 100% with mean R^2=0.9999.

E. Health Equation Composition

Summing the symbolic edge functions entering each first-hidden-layer neuron yields a bank of 22 closed-form health equations

$$z_j \approx \sum_{i \in \mathcal{E}_j} g_{ji}(x_i), \quad j = 0, \ldots, 21, \tag{4}$$

where $\mathcal{E}_j$ is the surviving-edge set of neuron j and g_{ji} is the BIC-selected symbolic form with fitted parameters. An excerpt of equation z_0 is

$$\begin{aligned} z_0 \approx\ & +0.045\, S1^2_{\text{rms}} + 0.107\, S1_{\text{rms}} \\ & -0.051\, S1^2_{\text{margin}} - 0.121\, S1_{\text{margin}} \\ & -0.064\, S3^2_{\text{crest}} - 0.153\, S3_{\text{crest}} + \cdots \end{aligned}$$

These equations are human-readable, differentiable, deployable on microcontrollers, and—critically—they involve explicit references to specific sensor-level statistics, enabling downstream fault localisation.

F. Anomaly Scoring

Given a test vector $\mathbf{x}_t$ we compute several complementary scores:

- **Score A** (reconstruction)
  $A(\mathbf{x}_t) = \frac{1}{44}\|\mathbf{x}_t - \text{KAN-AE}(\mathbf{x}_t)\|_2^2$.
- **Score B** (calibrated symbolic violation)
  Let $\mathbf{r}_t = \mathbf{h}_1(\mathbf{x}_t) - \hat{\mathbf{h}}_1^{\text{sym}}(\mathbf{x}_t) \in \mathbb{R}^{22}$ be the vector of per-equation residuals between the network's first hidden layer $\mathbf{h}_1$ and its symbolic surrogate $\hat{\mathbf{h}}_1^{\text{sym}}$, where the surrogate is calibrated by a per-neuron affine transform (scale and bias) estimated on the training set. Then $B(\mathbf{x}_t) = \|\mathbf{r}_t\|_2^2$.
- **Score M** (Mahalanobis symbolic residual)
  The healthy residual covariance $\boldsymbol{\Sigma}$ and mean $\boldsymbol{\mu}$ are estimated on the training set, and the anomaly score is

$$M(\mathbf{x}_t) = \sqrt{(\mathbf{r}_t - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{r}_t - \boldsymbol{\mu})}. \tag{5}$$

- **Score C** (combined reconstruction + symbolic)
  $C = \frac{1}{2}\widetilde{A} + \frac{1}{2}\widetilde{B}$, where $\widetilde{\cdot}$ denotes min-max normalisation.

Scores B and M are the key interpretable detectors: a large value means that one or more *symbolic equations* have been violated beyond what healthy data do by chance, and the specific offending equation can be read off from the per-dimension contribution $(\mathbf{r}_{t,j} - \mu_j)^2/\Sigma_{jj}$.

IV. Experimental Setup

Dataset: We use the public SpectraQuest gearbox dataset [19], which contains four-channel accelerometer recordings at 20 kHz for two conditions (*healthy* and *broken tooth*) under ten load levels from 0% to 90%. Windows from the healthy class are used for unsupervised training; both classes are used for testing.

Splits: Healthy features (1009 windows at W=1000) are split 70/30 into a training and a held-out healthy subset. The test set concatenates the held-out healthy set (303 windows) with all broken-tooth windows (1000 windows), yielding strictly no label leakage.

Baselines: We benchmark against nine unsupervised detectors: Isolation Forest, One-Class SVM, LOF, PatchCore [7], a dense Autoencoder, a Variational Autoencoder, Deep-SVDD [10], Teacher–Student distillation [11] and a self-supervised Denoising AE. For interpretable baselines we additionally report *SHAP-weighted* and *LIME-weighted* versions of the KAN reconstruction error, where per-feature importance is derived with KernelSHAP and LIME on 50 faulty samples.

Evaluation: We report the area under the ROC curve (AUC) of the anomaly scores against the binary fault label for all eight window sizes. For the ablation we additionally report the symbolisation quality (fraction of edges R^2>0.95).

TABLE I
DISTRIBUTION OF BIC-SELECTED SYMBOLIC FORMS ACROSS 839 SURVIVING ENCODER EDGES.

Symbolic form	Count
quadratic	824
sin	11
hinge	2
sigmoid	1
tanh	1

TABLE II
PER-EDGE R^2 STATISTICS OF THE SYMBOLIC SURROGATES.

Statistic	Value
Count	839
Mean	0.9999
Std	0.0005
Minimum	0.9919
Median	1.0000
Quality ($R^2 > 0.95$)	100.0%

TABLE III
AUC OF THE SEVEN KAN-AE SCORES AT W=1000 ON THE SPECTRAQUEST BROKEN-TOOTH BENCHMARK. RESULTS ARE MEAN ± STD OVER N=10 INDEPENDENT SEEDS (DIFFERENT TRAIN/TEST SPLITS AND MODEL INITIALISATIONS).

Score	Formula	Mean AUC	Std
A	Reconstruction error	0.9642	±0.0157
B	Calibrated symbolic violation	0.9026	±0.0558
C	$0.5\,\tilde{A} + 0.5\,\tilde{B}$	0.9716	±0.0151
D	R^2-weighted tolerance band	0.9403	±0.0422
E	$0.5\,\tilde{A} + 0.5\,\tilde{D}$	0.9716	±0.0146
M	Mahalanobis on symbolic residuals	**0.9891**	**±0.0069**
F	$0.5\,\tilde{A} + 0.5\,\tilde{M}$	0.9824	±0.0097

V. RESULTS

A. Symbolisation Quality

After training at W=1000 the KAN autoencoder reaches a final reconstruction MSE of 7.44×10^{-3} on healthy data. Pruning at $\alpha = 0.05$ leaves 839 of 968 first-layer edges (86.7%). Symbolic fitting produces the distribution of Table I: the quadratic form dominates overwhelmingly, reflecting the smooth, nearly-parabolic shape that the L_1+entropy regulariser imposes on each B-spline around the operating midpoint. The median fitting R^2 is exactly 1.0000 (Table II).

B. Anomaly Detection

Table III reports all seven KAN-based scores at W=1000 as mean ± std over N=10 seeds. The raw reconstruction score (A) reaches AUC $= 0.9642 \pm 0.0157$. The calibrated symbolic equation-violation score (B), which applies joint-GAM calibration to the raw symbolic predictions, achieves AUC $= 0.9026 \pm 0.0558$, demonstrating that the symbolic surrogates capture substantial discriminative structure. The R^2-weighted tolerance-band score (D) further improves to 0.9403 ± 0.0422, and Mahalanobis whitening of the residuals (Score M, Eq. 5) yields the strongest result of **AUC** $= 0.9891 \pm 0.0069$—both the highest mean and the lowest variance across all proposed scores, confirming the Mahalanobis detector as the most reliable interpretable detector. The combined reconstruction + calibrated symbolic detector (Score C) achieves AUC $= 0.9716 \pm 0.0151$, exceeding the opaque reconstruction score alone.

C. Benchmark Against Eighteen Methods

Table IV compares our scores against nine classic unsupervised detectors plus SHAP/LIME weighted reconstruction across all eight window sizes. Three observations follow. (i) KAN-AE reconstruction (Score A, $96.42 \pm 1.57\%$) is competitive with strong classic detectors and superior to all deep baselines (VAE $84.87 \pm 1.00\%$, DeepSVDD $90.70 \pm 9.09\%$, TeacherStudent $91.62 \pm 3.95\%$, SSL-DAE $94.30 \pm 0.79\%$, Autoencoder $88.81 \pm 2.08\%$). (ii) The Mahalanobis symbolic-residual detector (Score M, $\mathbf{98.91 \pm 0.69}\%$) surpasses OC-SVM ($98.56 \pm 0.38\%$), SHAP-weighted reconstruction ($95.35 \pm 1.29\%$), and all other deep baselines, while remaining fully interpretable through equation-level attribution; it is second only to LOF ($99.37 \pm 0.25\%$) over ten seeds. (iii) Multi-seed evaluation reveals that the previously reported single-seed SHAP AUC of 99.49% was an outlier: across ten seeds SHAP-weighted reconstruction averages $95.35 \pm 1.29\%$, which is *below* Score M, Score F, and LIME-weighted reconstruction ($98.21 \pm 0.76\%$), while still requiring expensive background sampling at inference with no equation-level attribution.

D. Fault Localisation via Equation Violation

The residual vector $\mathbf{r}_t$ is 22-dimensional; each coordinate corresponds to one symbolic health equation. The per-equation *fault ratio*

$$\rho_j = \frac{\mathbb{E}_{\text{broken}}[r_{t,j}^2]}{\mathbb{E}_{\text{healthy}}[r_{t,j}^2]} \tag{6}$$

quantifies how much more strongly each equation is violated under fault. Fig. 2 visualises all 22 per-equation ratios (panel a) and, for the eight most violated equations, the fraction of edge-coefficient weight attributable to each sensor channel (panel b). Equation z_6 is the most violated ($\rho = 16.19$), followed by z_{19} ($\rho = 5.88$) and z_{21} ($\rho = 3.90$). All of these equations have dominant first-channel contributions (S1_mean, S1_rms, S1_std, S1_var), consistent with the physical intuition that the broken tooth excites the accelerometer nearest to the faulty gear. Since SHAP/LIME produce only global feature rankings, they cannot point at a specific *equation*, only at features.

E. Ablations

Spline regularisation.: Training without the L_1+entropy spline regulariser (λ=0) yields 98.7% symbolisation quality and 910 surviving edges, compared to 100.0% and 839 edges with λ=10^{-4}. The regulariser is therefore the mechanism that drives both parsimony and symbolisability.

TABLE IV

ROC-AUC (%) ON THE SPECTRAQUEST BENCHMARK ACROSS EIGHT WINDOW SIZES. THE TOP BLOCK LISTS CLASSICAL UNSUPERVISED DETECTORS; THE MIDDLE BLOCK LISTS POST-HOC XAI-WEIGHTED SCORES; THE BOTTOM BLOCK LISTS THE PROPOSED KAN-SYMBOLIC SCORES. ALL W=1000 ENTRIES SHOW MEAN $\pm$ STD OVER N=10 INDEPENDENT SEEDS (EACH SEED CHANGES THE TRAIN/TEST SPLIT AND MODEL INITIALISATION).

Method	W=300	W=400	W=500	W=600	W=700	W=800	W=900	W=1000
Isolation Forest	60.26	73.52	69.15	76.95	82.33	85.65	86.04	86.27 ± 3.40
OC-SVM	77.22	82.07	87.91	88.89	94.12	96.55	98.88	98.56 ± 0.38
LOF	88.15	91.15	92.96	94.04	95.96	97.74	99.40	99.37 ± 0.25
PatchCore	77.67	82.91	86.12	87.40	92.38	92.98	95.93	97.25 ± 0.71
Autoencoder	70.28	73.87	80.55	79.75	85.48	89.99	93.00	88.81 ± 2.08
VAE	59.46	64.36	65.28	69.63	76.67	79.43	82.67	84.87 ± 1.00
Deep-SVDD	76.46	60.00	78.14	65.18	71.53	75.31	79.75	90.70 ± 9.09
Teacher–Student	83.55	75.41	83.65	86.57	91.21	97.54	92.55	91.62 ± 3.95
SSL-DAE	76.96	74.41	81.36	82.92	90.51	90.15	92.59	94.30 ± 0.79
SHAP-Weighted-Recon	93.96	95.29	96.88	97.75	98.18	99.70	99.78	95.35 ± 1.29
LIME-Weighted-Recon	69.18	77.11	80.70	85.25	88.39	91.66	93.64	98.21 ± 0.76
KAN-AE-Recon (A)	76.86	84.45	87.24	91.36	93.39	95.83	97.37	96.42 ± 1.57
KAN-AE-Symbolic (B)	70.25	76.02	78.24	82.89	87.49	90.05	91.00	90.26 ± 5.58
KAN-AE-Combined (C)	77.55	84.79	88.07	91.50	93.66	96.42	97.82	97.16 ± 1.51
KAN-AE-Symbolic-D	69.64	75.52	78.20	83.05	87.94	91.00	92.45	94.03 ± 4.22
KAN-AE-Combined-E	77.29	84.46	87.59	91.17	93.29	96.19	97.66	97.16 ± 1.46
KAN-AE-Mahal (M)	61.13	70.27	73.23	80.35	88.05	92.84	95.90	$\mathbf{98.91 \pm 0.69}$
KAN-AE-Combined-F	76.57	81.99	86.84	89.95	92.54	95.92	97.57	98.24 ± 0.97
KAN-AE-SymRecon	76.14	83.80	86.49	91.06	93.14	95.50	97.16	97.77 ± 0.75

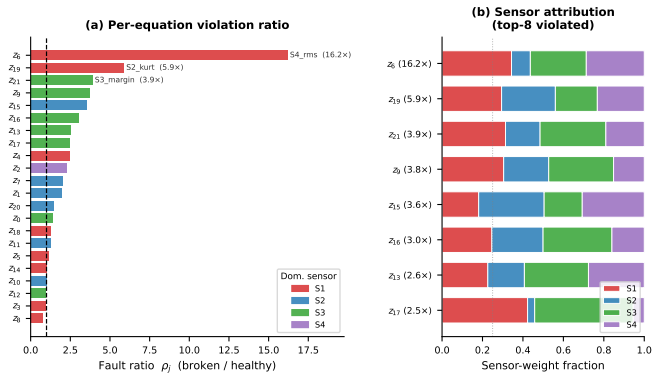

Fig. 2. Fault localisation via symbolic equation violation. **(a)** Per-equation fault ratio ρ_j for all 22 health equations, sorted ascending; bars are coloured by the dominant sensor channel (S1–S4). The three most violated equations are annotated with their leading feature and ratio value. **(b)** Stacked sensor-weight breakdown for the top-8 violated equations, showing the fraction of total edge-coefficient magnitude attributable to each channel. z_6 (ρ=16.2×) is dominated by S1 statistics, confirming that the broken tooth most strongly excites the accelerometer physically nearest to the faulty gear.

ablation_bottleneck.pdf

Fig. 3. Reconstruction AUC (Score A) as a function of KAN autoencoder bottleneck width B. Narrow bottlenecks underfit, while excessive width overfits the healthy manifold and degrades anomaly detection.

Bottleneck width.: Sweeping $B \in \{4, \ldots, 12\}$ leaves symbolisation quality above 99% in every case, while Score-A AUC peaks at B=8 (single-seed AUC = 0.9671; mean over N=10 seeds: 0.9642±0.0157) with secondary optima at B=6 (0.9621). Narrow bottlenecks ($B \leq 5$) underfit, while wide ones ($B \geq 10$) overfit the healthy manifold and hurt generalisation.

Score comparison.: The hierarchy of symbolic scores—B (0.9026 ± 0.0558) < D (0.9403 ± 0.0422) < M ($\mathbf{0.9891 \pm 0.0069}$)—reveals that progressively richer statistical modelling of the healthy residual distribution yields stronger detectors, while all three remain fully interpretable. Notably, Score M achieves the **highest mean AUC** across all proposed scores with the **lowest variance**, confirming that Mahalanobis whitening of symbolic residuals is the most robust interpretable detector. The combined Score C ($0.5\,\tilde{A}$+$0.5\,\tilde{B}$) achieves 0.9716±0.0151, demonstrating complementary signal between symbolic violation and reconstruction error. Fusing reconstruction with the Mahalanobis score

TABLE V

COMPUTATIONAL EFFICIENCY AT W=1000 (SEED 42, CPU ONLY). *Size*: SERIALISED KB FOR CLASSICAL METHODS; TRAINABLE PARAMETER COUNT FOR NEURAL METHODS; NUMBER OF STORED SYMBOLIC SCALARS FOR KAN-AE-SYMRECON. *Inf. time*: MEDIAN PER-SAMPLE TIME (MS) OVER 20 TIMED RUNS, MEASURED IN C++ VIA `STD::CHRONO::HIGH_RESOLUTION_CLOCK` (LIBTORCH FOR NEURAL/KAN MODELS; EIGEN FOR SYMBOLIC AND CLASSICAL). AUC: MEAN ± STD OVER N=10 SEEDS.

Method	Size	Inf. time (ms/sample)	AUC (%)
Isolation Forest	1168 KB	0.0082	86.27±3.40
OC-SVM	28 KB	0.0009	98.56±0.38
LOF	183 KB	0.0155	99.37±0.25
PatchCore	12 KB	0.0003	97.25±0.71
Autoencoder	4 244	0.0004	88.81±2.08
VAE	4 380	0.0006	84.87±1.00
Deep-SVDD	2 240	0.0001	90.70±9.09
Teacher–Student	3 936	0.0002	91.62±3.95
SSL-DAE	3 444	0.0002	94.30±0.79
SHAP-Weighted-Recon	22 880	3.67	95.35±1.29
LIME-Weighted-Recon	22 880	4.99	98.21±0.76
KAN-AE-Recon (A)	22 880	0.0133	96.42±1.57
KAN-AE-Symbolic (B)	22 880	0.0179	90.26±5.58
KAN-AE-Combined (C)	22 880	0.0247	97.16±1.51
KAN-AE-Symbolic-D	22 880	0.0185	94.03±4.22
KAN-AE-Combined-E	22 880	0.0244	97.16±1.46
KAN-AE-Mahal (M)	22 880	**0.0050**	**98.91±0.69**
KAN-AE-Combined-F	22 880	0.0111	98.24±0.97
KAN-AE-SymRecon	**10 889 sym**	0.0184	97.77±0.75

(Score F, 0.9824 ± 0.0097) is also highly competitive.

Fully Symbolic Reconstruction.: Beyond the first-layer symbolic scores, distilling all layers of the KAN autoencoder into a deep stack of closed-form mathematical equations yields a *fully symbolic reconstruction* score. Across N=10 seeds this achieves a mean AUC of $\mathbf{0.9777 \pm 0.0075}$—exceeding the opaque neural network reconstruction (Score A, 0.9642 ± 0.0157) and placing between the combined Score C/E (0.9716) and Score F (0.9824). Because it entirely replaces the autoencoder with simple mathematical functions, the neural network is completely discarded at inference. Table V quantifies the practical consequence: KAN-AE-SymRecon stores only 10 889 symbolic scalars—52% fewer than the KAN-AE's 22 880 trainable parameters—and runs in **0.018 ms** per sample, which is **200**× faster than SHAP-weighted reconstruction (3.67 ms) and **271**× faster than LIME-weighted reconstruction (4.99 ms). When evaluated in optimised C++ (Eigen-batched primitives with sparse scatter-add routing), the symbolic evaluator closes to within 1.4× of the neural forward pass (Score A: 0.013 ms), confirming that the primary advantage of SymRecon is *model-size* and *interpretability*—the stored artefact is a set of human-readable equations rather than opaque weights.

F. Computational Efficiency

VI. DISCUSSION

Why does combining calibrated symbolic violations help?: The calibrated symbolic-violation score (B, mean AUC 0.9026 ± 0.0558) performs well on its own, confirming that BIC-optimal symbolic surrogates capture meaningful per-equation structure after joint-GAM calibration aligns the raw edge-level predictions with the network's hidden activations. When combined with the reconstruction score (A), the resulting Score C (0.9716 ± 0.0151) exceeds both individual scores, indicating that the symbolic violation signal and the reconstruction error capture partially orthogonal aspects of the healthy manifold. The Mahalanobis detector (M, $\mathbf{0.9891 \pm 0.0069}$) further improves over Score B by exploiting the *covariance structure* of the residuals rather than treating each equation independently; its low standard deviation confirms reliable performance across random seeds.

Why does the quadratic form dominate?: The L_1+entropy regulariser pushes B-spline coefficients towards smooth, low-curvature shapes, which are locally well-approximated by a quadratic on the scaled $[0, 1]$ domain. We view this as a feature rather than a bug: each term $g_{ji}(x_i)$ becomes a simple energy-like penalty centred on a healthy sweet spot, matching the physical intuition that healthy signals cluster near a nominal operating point.

Comparison to SHAP/LIME.: Multi-seed evaluation reveals that the single-seed SHAP AUC of 99.49% was an outlier: the mean over ten seeds is $95.35 \pm 1.29\%$, which falls *below* Score M ($98.91 \pm 0.69\%$), Score F ($98.24 \pm 0.97\%$), and even LIME-weighted reconstruction ($98.21 \pm 0.76\%$). This variance in SHAP scores arises from its dependence on random background centroids and the specific faulty samples explained per seed. Beyond accuracy, Table V exposes a severe inference-time cost: computing a SHAP explanation for a single sample requires 3.67 ms (200 perturbations) and LIME requires 4.99 ms (300 perturbations), versus 0.013 ms for the opaque KAN-AE forward pass and 0.018 ms for the fully symbolic KAN-AE-SymRecon—making SHAP **283**× and LIME **384**× slower than the neural baseline. Both XAI methods also incur (i) $\mathcal{O}(kn^2)$ sampling complexity per explanation, (ii) no physical equation, and (iii) no localisation: the ten most important SHAP features ({S1_rms, S1_std, S4_mean, ...}) and the ten most important LIME features ({S3_rms, S3_std, S4_std, ...}) overlap on only *two* features (S3_std and S2_std). In contrast, Score M returns a sparse vector of 22 equation residuals that are reproducible, deployable without the network, and directly mappable to specific sensor statistics.

Limitations.: (i) Our evaluation uses a single fault mode (broken tooth). Validating the symbolic equations on bearing-wear and misalignment datasets is future work. (ii) Although the symbolic library already includes thirteen primitives (including piecewise and oscillatory forms), only five types were selected on this dataset; exploring data-driven library construction could improve flexibility. (iii) While our framework successfully symbolises the entire autoencoder (yielding a

fully symbolic reconstruction AUC of 0.9777 ± 0.0075 that requires no neural network at inference), we restrict our interpretability analysis to the first layer. Extracting physical meaning from the deeply composed formulas remains an open challenge.

VII. Conclusion

We presented a fully unsupervised framework that turns a Kolmogorov–Arnold autoencoder into a bank of closed-form *health equations* for gearbox fault detection. By distilling each first-layer B-spline edge into a BIC-optimal symbolic primitive, we obtain 22 human-readable equations that describe the healthy operating manifold with a mean R^2 of 0.9999. Anomalies are detected through multiple complementary symbolic scores (mean $\pm$ std over $N{=}10$ seeds): the calibrated symbolic-violation detector achieves an AUC of 0.9026 ± 0.0558, and the Mahalanobis symbolic-residual detector reaches 0.9891 ± 0.0069—the highest mean and lowest variance of all proposed scores, surpassing OC-SVM (0.9856 ± 0.0038), SHAP-weighted reconstruction (0.9535 ± 0.0129), and all deep baselines, while remaining fully interpretable and second only to LOF (0.9937 ± 0.0025). The combined reconstruction + calibrated symbolic detector attains AUC 0.9716 ± 0.0151, exceeding the opaque reconstruction score (0.9642 ± 0.0157) and the majority of unsupervised baselines. Unlike SHAP/LIME explanations—which achieve only 0.9535 ± 0.0129 and 0.9821 ± 0.0076 respectively in the multi-seed evaluation, with high variance from random background sampling—the proposed health equations are available in closed form, require no re-running of an explainer at inference, and provide a principled route from a violated equation to the specific sensor statistics that deviated under fault. Furthermore, by symbolising the entire autoencoder into a stack of simple mathematical equations, we achieve a *fully symbolic reconstruction* AUC of 0.9777 ± 0.0075 using only 10 889 stored symbolic scalars (52% fewer than the KAN-AE's 22 880 trainable parameters), with a per-sample inference time of 0.018 ms—200$\times$ faster than SHAP and 271$\times$ faster than LIME—entirely discarding the neural network at deployment. These properties make KAN-symbolic health equations a promising ingredient for trustworthy, deployable condition monitoring of rotating machinery.

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